Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Main Topic: [Banana] – [disease management]

1.1 Panama Wilt / Fusarium Wilt (*Fusarium oxysporum* f. sp. *cubense*)

Causal Agent and Susceptible Cultivars Panama Wilt (Fusarium Wilt) is a lethal, soil-borne fungal disease caused by *Fusarium oxysporum* f. sp. *cubense*. In India, Races 1, 2, and 4 have been reported to attack most banana varieties, including Cavendish cultivars. Susceptible varieties identified across the available sources include Rasthali, Monthan, Virupakshi, Karpuravalli, Ney Poovan, and Pachanadan. Cultivars listed as tolerant or resistant include Karpuravalli, Monthan, Nattupoovan, and Kunnan.

Symptoms and Identification Disease symptoms are most pronounced on plants that are at least 5 months old, although under highly favorable conditions, 2-to-3-month-old plants can also succumb.

- **Foliar Symptoms:** The earliest visual sign is faint yellowing or chlorotic streaking along the petioles and leaf blades of the lower, oldest leaves, progressing from the leaf margin toward the midrib and advancing to upper leaves.
- **Leaf Collapse:** Leaves exhibit drooping and break near their base, hanging down around the pseudostem to form a characteristic "skirt-like" appearance. In the yellowing symptom type, leaves gradually yellow and collapse; in the non-yellowing type, leaf petioles collapse without prior chlorosis. Often, all leaves except the youngest collapse, leaving only the central heart leaf standing upright.
- **Pseudostem Splitting:** Conspicuous longitudinal splitting of the outer leaf sheaths of the pseudostem occurs near the ground.
- **Internal Vascular Discoloration:** Vascular vessels in the pseudostem and corm show continuous yellowish, reddish, or dark brown streaks, with color intensification concentrated toward the rhizome/corm periphery. Wilting results from the physical blocking of xylem and phloem transport.
- **Terminal Decline:** Within 4 to 6 weeks after petiole streaks appear, the plant dies, leaving only a dead, standing pseudostem trunk.

Pathogen Cycle, Primary Inoculum, and Spread

- **Primary Inoculum and Survival:** The fungus is a soil-borne organism that survives in soil or infected rhizomes/suckers primarily as resting chlamydospores formed by hyphal and conidial cells. In the absence of a host, it persists in previously colonized plant tissue in the soil.

- **Entry and Primary Spread:** Infection occurs through root injuries or natural root entry points. Primary spread to non-infested fields occurs when symptomless, infected suckers or corms are unwittingly planted.
- **Secondary Spread:** Microconidia and macroconidia spread locally via irrigation water, surface runoff, soil adhering to farm implements/machinery, and directly through root contacts in interconnected root mats. Continuous monocropping accelerates inoculum buildup in the soil.

Favorable Environmental Conditions and Aggravating Factors The disease develops severely under the following conditions:

- Warm soil temperatures, acidic soils, light-textured soils, or poorly drained heavy soils.
- Low soil moisture, flooding irrigation, and monocropping of susceptible cultivars.
- Root damage caused by intercultural operations or infestation by plant-parasitic nematodes (such as *Radopholus similis*), which predispose root tissues to fungal entry.

Damage Panama Wilt causes complete vascular blockage, stunting, severe canopy defoliation, and plant death. Infested fields suffer severe yield loss and long-term soil contamination, rendering land unsuitable for banana cultivation for extended periods.

Prevention and Integrated Disease Management

1. Exclusion, Quarantine, and Sanitation

- Restrict movement of infected suckers or soil from infested areas to clean fields.
- Promptly uproot diseased plants along with the surrounding soil, remove them from the field, and burn or destroy them.
- Disinfect all farming tools and implements with spirit or formaldehyde after working in infested areas before entering disease-free plots.
- Apply quick lime or agricultural lime @ 1–2 kg per pit (or 3 kg per pit) to the soil after uprooting infected plants.

2. Crop Rotation, Soil Inundation, and Bio-Disinfection

- **Crop Rotation:** Highly infected fields should not be planted with banana for at least 3 to 5 years. Recommended rotation schemes include wetland paddy or sugarcane for 2–3 cycles, as well as tapioca, onion, or pineapple.
- *Source Disagreement Note:* While *IPM-Banana-Revised-Sept2011.pdf* (p. 9) and *Banana_disease_detection.pdf* (p. 31) explicitly recommend sugarcane in crop rotations to reduce wilt, *Banana_disease_1.pdf* (p. 2) cautions to avoid sugarcane or sunflower in crop rotation.

- **Soil Inundation and Flooding:** Inundate/flood the field with water for 1 to 3 months prior to planting. Flood fallowing for 5 months destroys *Fusarium* species in the soil.
- **Biological Soil Disinfection:** Spread 500–1,000 kg/acre of paddy or maize straw and flood the field for 20–30 days. Alternatively, broadcast Sunn hemp seeds, grow for 45 days, and perform *in situ* ploughing.

3. Healthy Planting Material and Corm Treatment

- Use certified disease-free suckers or tissue-cultured banana plants (with 5–6 leaves).
- **Corm Paring and Dipping:** Pare/trim corms by removing roots and outer skin until clean tissue is exposed. Dip pared corms in one of the following solutions before planting:
  - 0.1% Emisan solution (1 g in 1 L water) for 5 minutes.
  - Carbendazim solution @ 10 g / 10 L water (0.2%).
  - Carbendazim (2 g/L) + 0.05% monocrotophos for 30 minutes.
  - Alternatively, perform pralinage with 40 g Carbofuran 3G granules per sucker, or dip corms in 0.75% Monocrotophos and shade-dry for 24 hours prior to planting.

4. Prophylactic Chemical Drenching and Corm Injection

- **Soil Drenching:** Drench soil around plants with 0.2% Carbendazim solution as a prophylactic measure (starting bimonthly from 5 or 6 months after planting).
- **Corm Injection / Capsule Insertion:** Inject 3 mL of a 2% Carbendazim solution (20 g/L) into the corm, or embed a 50 mg gelatin capsule of Carbendazim into the corm at the 5th, 7th, and 9th months after planting.

5. Biological Control and Organic Amendments

- **Bioagents:** Apply powder formulations of *Trichoderma viride*, *Trichoderma harzianum*, *Pseudomonas fluorescens*, or *Bacillus subtilis* @ 15 g to 30 g per plant mixed with 1–10 kg FYM/compost. Apply 4 doses: at planting pit, and subsequently at the 3rd, 5th, and 7th months after planting.
- **Chaffy Grain Formulation:** Soil application of *Trichoderma viride* chaffy grain formulation @ 20 g/plant (2×10^8 cfu/g) along with FYM at planting and at 2-month intervals reduces Fusarium wilt incidence by up to 80% in tissue-cultured and sucker-derived plants (e.g., cv. Rasthali).
- **Organic Amendments:** Apply neem cake @ 250 kg/ha or 300 g/plant at the 5th month after planting.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Diseases – Panama Wilt, pp. 1–2).
2. *Banana Expert System* (URL; Sections: Cropping System, Do's and Don'ts, Overcoming Wilt Disease, Monthly Schedule).
3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.1: Panama wilt / Fusarium wilt, pp. 30–31).
4. *IPM-Banana-Revised-Sept2011.pdf* (Extension Bulletin No. 3, National Horticulture Mission 2012; Section 9: Panama wilt, pp. 8–9; Section 15: IPM/INM Schedule, p. 18).

1.2 Sigatoka Leaf Spot Complex (Yellow, Black, and Septoria/Eumusae Leaf Spot)

Causal Agents and Species Complex Sigatoka leaf spot is a complex of destructive fungal diseases affecting banana foliage. Across the available sources, three primary fungal species are identified as causal agents:

- **Yellow Sigatoka:** Caused by *Mycosphaerella musicola*.
- **Black Sigatoka / Black Leaf Streak:** Caused by *Mycosphaerella fijiensis* (and reported as *Mycosphaerella fijiensis* var. *difformis*).
- **Eumusae / Septoria Leaf Spot:** Caused by *Mycosphaerella eumusae*.

Symptoms and Disease Identification Symptoms vary slightly by specific causal agent and disease stage, but follow a characteristic progression:

- **Initial Stage:** Faint light green or yellowish narrow specks (approximately 1 mm in length) appear on the upper surface of older leaves.
- **Spot Enlargement:** Specks enlarge into linear, oblong, or oval spots. Their color transitions from light yellow/brownish-green to dark brown or black, surrounded by a dark brown band and a yellow halo.
- **Fruiting Bodies and Necrotic Centers:** Black specks of fungal fructification appear within affected leaf tissue. Over time, the center of the spot dies, turning light grey with a surrounding brown border.
- **Coalescence and Leaf Blight:** Severe infections cause numerous spots to coalesce, resulting in extensive leaf tissue necrosis, rapid drying, and premature defoliation starting from the leaf tip.

Disease Development, Spread, and Favorable Conditions

- **Inoculum Survival and Dissemination:** The pathogen survives in infected banana leaves, trash, and plant debris. Primary and secondary spread occurs via wind-borne ascospores and conidia, rain splashes, water droplets, transport of infected leaves/plants, and using diseased foliage as wrapping or packing material for banana bunches.
- **Environmental Triggers:** High rainfall, persistent relative humidity, heavy dew, surface leaf wetness, and warm temperatures (above 21°C) favor mass infection, spore release, and rapid development. Outbreaks are particularly severe during rainy seasons and floods.
- **Agronomic and Soil Aggravating Factors:** Poor soil drainage, low soil fertility, weed infestation, continuous year-round banana monocropping, lack of crop rotation, and failure to apply regular fungicidal sprays accelerate disease severity.
- **Spacing and Canopy Microclimate Effects:** High-density planting (e.g., 1.25 m× 1.25 m or paired-row planting) increases relative humidity and leaf wetness duration.
- *Source Observation on Plant Spacing: IPM-Banana-Revised-Sept2011.pdf* (p. 13) highlights closer spacing (1.25 m × 1.25 m) as a major driver of disease severity and recommends 1.5 m × 1.5 m. Meanwhile, *Banana Expert System* (Section: Diseases incidence) notes regional nuances: wider spacing (2.4 m × 2.4 m) reduced disease in 2nd ratoon Rajapuri in North Karnataka, whereas in Trichy (Tamil Nadu), dense hill planting (3 suckers per hill) maintained lower relative humidity and disease when combined with wider inter-hill spacing.

Yield Loss and Commercial Damage Sigatoka inflicts severe economic damage across commercial cultivars:

- **Leaf Area Reduction:** To produce a normal commercial yield, banana plants require 17–20 functional green leaves during the vegetative phase (6 months), 12–15 green leaves during shooting/flowering, and 7–10 leaves at harvest. Severe Sigatoka defoliation often leaves plants with fewer than 10 functional leaves (or as few as 15–18 total required leaves severely reduced).
- **Fruit Development Effects:** Impaired photosynthesis delays flowering and causes reduced hand/finger count, small and poorly filled fingers, and finger splitting.
- **Export and Transit Losses:** Infected orchards produce fruit with drastically shortened shelf life, premature ripening in transit, and peel defects, making bunches unsuitable for export.

Integrated Disease Management (IDM)

1. Cultural Practices, Field Sanitation, and Schedule

- **Planting Window and Spacing:** Plant strictly during recommended months (June and October) using proper spacing (1.5 m × 1.5 m).
- **Deleafing and Sanitation:** Inspect fields regularly. Remove dried or infected leaves once a month starting from the 2nd month after planting. Destroy removed foliage by burning outside the field or burying it in manure pits covered with soil.
- **Desuckering and Drainage:** Prune surplus side suckers regularly to prevent canopy overcrowding and maintain adequate field aeration. Improve field drainage to reduce humidity. Keep fields weed-free.
- **Balanced Nutrition:** Apply recommended fertilizer doses based on soil testing to avoid excessive succulent vegetative growth.

2. Biological Control and Prophylactic Treatments

- **Bio-fungicides:** Apply foliar sprays of *Bacillus subtilis* 1.50% LF (T stanes BS-1 strain MTCC 25072) @ 5 L/ha in 750 L of water.
- **Flood Prophylaxis:** During flood conditions, apply prophylactic sprays of 0.1% Propiconazole, 0.2% Carbendazim, and 0.25% Mancozeb at 15-day intervals between sprays.

3. Mineral Oil Emulsions and Fungicidal Spray Schedule Combining fungicides with biodegradable mineral spray oil (e.g., Banole® oil @ 1%) enhances cuticle penetration, provides a direct fungistatic film that slows fungal growth, and improves active ingredient distribution.

- **Oil-Fungicide Emulsion Schedule:** Spray Banole oil (1%) mixed with half-dose systemic/contact fungicides at 15–20 day intervals, starting from the 2nd month after planting:

  1. *1st Spray:* Banole oil 1% + Carbendazim @ 0.5 g/L.
  2. *2nd Spray:* Banole oil 1% + Companion @ 0.5 g/L.
  3. *3rd Spray:* Banole oil 1% + Carbendazim @ 0.5 g/L + Calixin @ 0.5 mL/L.
  4. *4th Spray:* Banole oil 1% + Propiconazole @ 0.5 mL/L.
  5. *5th Spray:* Banole oil 1% + Carbendazim @ 0.5 g/L.
  6. *6th Spray:* Banole oil 1% + Propiconazole @ 0.5 mL/L.
  7. *7th Spray:* Banole oil 1% + Carbendazim @ 0.5 g/L.
  8. *8th Spray:* Banole oil 1% + Propiconazole @ 0.5 mL/L.

- **Fungicide Resistance Management:** Alternate systemic and contact fungicides at 7–10 day intervals to prevent resistant strain buildup.
- **Seasonal Spray Recommendation:** In winter/cool months, spray 0.1% Propiconazole (Tilt) covering both upper and lower leaf surfaces with a wetting agent.

4. Recommended Chemical Fungicides and CIB&RC Approved Dosages

- **Copper Oxychloride 50% WP:** Apply @ 2.5 g/L (1.25 kg a.i. / 2.5 kg formulation in 750–1000 L water/ha) or 3 g/L.
- **Mancozeb 75% WP / Dithane M-45:** Apply @ 1.5–2 kg formulation (1.125–1.5 kg a.i.) in 750 L water/ha.
- **Thiophanate Methyl:** Apply @ 1 g/L water.
- **Propiconazole:** Apply @ 1 mL/L water for 3 to 5 sprays at 25–30 day intervals.
- **Pyraclostrobin 20% WG:** Apply @ 750 g formulation (150 g a.i.) in 500 L water/ha (Pre-Harvest Interval [PHI]: 103 days).
- **Metiram 55% + Pyraclostrobin 5% WG:** Apply @ 1500–1750 g formulation (900–1050 g a.i.) in 500 L water/ha (PHI: 85 days).
- **Pyraclostrobin 133 g/L + Epoxiconazole 50 g/L SE:** Apply @ 750 mL formulation (137.25 g a.i.) in 500 L water/ha (PHI: 24 days).

Sources Cited

1. *Banana_disease_1.pdf* (Section: Leaf Spot, Leaf Streak or Sigatoka Disease, p. 2).
2. *Banana Expert System* (URL; Sections: Diseases incidence, Sixth Month, Overcoming flood damage).
3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.2: Eumusae Leaf Spot Disease, pp. 31–32; Section 18: CIB&RC approved pesticides, pp. 44–45).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 8: Sigatoka Leaf Spot Disease, pp. 12–14; Section 15: IPM/INM Schedule, p. 18).

1.3 Banana Bunchy Top Virus (BBTV)

Causal Agent, Viral Structure, and Vector Banana Bunchy Top Virus (BBTV) is a severe viral pathogen affecting banana crops worldwide. While historically hypothesized to be a luteovirus or double-stranded (ds) RNA virus, molecular studies indicate that a single-stranded DNA (ssDNA) virus causes or plays a primary role in disease development, potentially with an RNA virus co-involved.

- **Insect Vector:** The virus is transmitted exclusively by the banana black aphid, *Pentalonia nigronervosa* Coquerel (Hemiptera: Aphididae).
- *Source Disagreement Note on Transmission Mode: Banana_disease_detection.pdf* (p. 36) describes vector transmission as occurring in a **persistent manner**, whereas *IPM-Banana-Revised-Sept2011.pdf* (p. 15) states that transmission occurs in a **semi-persistent manner**.
- **Varietal Susceptibility:** Dwarf banana cultivars are reported to be particularly susceptible, although the virus infects most commercial cultivars (including Nendran, Robusta, Poovan, Ney Poovan, Rasthali, Red Banana, Monthan, and Karpuravalli).

Symptoms and Identification Symptoms vary depending on whether infection originates from infected planting material or vector transmission:

- **Diagnostic Foliar Vascular Symptoms:** The most characteristic diagnostic symptom is the presence of dark green "dot-and-dash" or interrupted dark green streaks along the secondary veins on the ventral surface of the leaf lamina, as well as along the midrib and petiole.
- **Foliar Deformities:** Newly emerged leaves are narrow, chlorotic, stunted, and exhibit mosaic streaking. Leaves become pale, stiff, and upright with brittle texture and upwards-rolled margins.
- **Rosette / "Bunchy Top" Canopy:** As successive leaves emerge severely stunted with short petioles, they crowd together at the apex of the pseudostem, forming a compact rosette or "bunchy top" appearance.
- **Symptom Latency:** Symptomless infection or viral latency can persist in plants or suckers for up to two full seasons before visual expression.
- **Fruit and Bunch Effects:** Primary infection in early stages causes extreme stunting; affected plants fail to flower or produce bunches. Late-stage secondary infection may allow a bunch to emerge, but fingers remain small, distorted, malformed, and fail to ripen or mature.

Epidemiology, Inoculum Sources, and Spread

- **Primary Spread:** Occurs through the movement and planting of symptomless, infected suckers, corm bits, or micropropagated tissue-culture plantlets derived from infected mother mats.
- **Secondary Local Spread:** Transmitted locally from plant to plant by colonies of *Pentalonia nigronervosa* feeding in leaf axils, beneath petiole bases, and on young suckers.
- **Seasonal Vector Activity:** Secondary spread peaks during winter in the plains and during summer in hill banana regions due to heightened aphid population dynamics and mobility.

Damage and Economic Loss BBTV causes complete loss of commercial yield on affected mats, severe stunting, and long-term orchard contamination. Because symptomless latent suckers harbor the virus, unmanaged fields become perpetual reservoirs that degrade regional production.

Integrated Disease Management (IDM)

1. Exclusion, Domestic Quarantine, and Certification

- Enforce strict legal and domestic quarantine regulations to prevent the movement of planting material or suckers from endemic areas (especially hill banana regions) into disease-free zones.

- Plant exclusively certified, virus-indexed tissue-cultured plants or clean suckers sourced from certified, vector-proof mother nurseries. Virus indexing via ELISA or DIBA using reliable antiserum is required for micropropagation explants.

2. Field Sanitation, Rouging, and In Situ Eradication

- Maintain regular orchard inspection schedules to detect early viral symptoms.
- **Systematic Eradication (Rouging):** Uproot affected plants along with the entire mat, rhizome system, and lateral suckers. Chop the dug-out plant tissue into small pieces, allow to dry, and burn or incinerate completely outside the plantation.
- **Chemical Plant Destruction:** Inject diseased plants *in situ* with kerosene or systemic weedicides/herbicides such as 2,4-D or 2,4,5-T. Dig out and chop the dead corm into small sections and re-treat to prevent sucker regrowth that could host the virus.
- **Community Eradication and Crop Holding:** In plantations where BBTV incidence exceeds 50%, adopt a community-wide eradication protocol and observe a non-host crop holding/fallow period for 2 to 3 seasons.

3. Vector (Aphid) Suppression and Chemical Control

- *Note on Vector Control Limits: IPM-Banana-Revised-Sept2011.pdf* (p. 16) emphasizes that applying insecticides to suppress aphids does not effectively stop BBTV spread unless combined with immediate rouging of infected plants.
- **Insecticidal Foliar Sprays:** Spray infected plants (prior to rouging) and all surrounding healthy mats with systemic insecticides:
  - Metasystox @ 0.1–0.5%.
  - Dimethoate 30 EC @ 1–2 mL/L.
  - Methyl demeton / Oxy-demeton methyl 25 EC @ 2 mL/L.
  - Imidacloprid @ 0.50 mL/L applied as a foliar spray at monthly intervals when aphid colonies build up.
  - Neemkosol @ 2.5 mL/L or Monocrotophos @ 2 mL/L.
- **Soil Granular Application:** Apply Carbofuran 3G @ 166 g/plant around the root zone.
- **Biological Vector Control:** Apply foliar sprays of the entomopathogenic fungus *Verticillium lecanii* @ 20 g/L (or 5 mL/L) at 25, 65, and 165 days after planting.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Banana Bunchy Top Virus, p. 3).
2. *Banana Expert System* (URL; Section: Do's and Don'ts – Removal of infected plants).

3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 12.3: Banana Aphid, pp. 21–22; Section 13.6: Bunchy top, p. 36).

4. *Banana_pests_3.pdf* (Section: Banana Aphid, p. 3).

5. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 3: Banana aphid, p. 4; Section 15: Bunchy Top, pp. 15–17).

1.4 Erwinia Rhizome Rot / Tip Over / Head Rot (*Erwinia carotovora* subsp. *carotovora*)

Causal Agent and Taxonomy Erwinia Rhizome Rot (also referred to as Tip Over or Head Rot) is a bacterial disease caused primarily by *Erwinia carotovora* subsp. *carotovora* (also cited as *Erwinia carotovora* / *Erwinia chrysanthemi*). The pathogen is a soil-borne, Gram-negative, rod-shaped bacterium with peritrichous flagella that exists individually or in pairs and chains.

Symptoms and Identification The disease attacks plants at various growth stages, but symptoms are particularly severe on young suckers:

- **Sucker Rotting and Foul Odor:** Newly planted suckers and young offshoots show soft rotting of the corm and collar region, emitting a characteristic foul, putrid odor.
- **Foliar Epinasty and Sudden Drying:** Rotting at the collar region leads to epinasty (downward curling) of leaves, followed by sudden leaf yellowing, desiccation, and death.
- **Pseudostem Detachment ("Tip Over"):** Because the collar tissues decay completely, affected pseudostems lose structural attachment to the corm. The plant easily tips over or pulls out cleanly from the collar region upon slight pressure or hand pulling, leaving the corm and roots behind in the soil.
- **Bacterial Ooze:** Bacterial slime or ooze is visibly exudated at the cut edges or interface between the corm and pseudostem, serving as a confirmatory field diagnostic sign.
- **Older Plant Symptoms:** In established or mature plants, rotting occurs around the trunk base and lower leaf bases. The trunk base becomes characteristically swollen, spongy, and split.

Disease Cycle, Primary Inoculum, and Spread

- **Primary Inoculum and Entry:** The bacterium is soil-borne and survives in infected plant debris and soil. Entry into host tissues occurs through natural leaf sheath openings or physical wounds on suckers and roots.
- **Secondary Spread:** Bacteria spread locally via contaminated irrigation water, surface runoff, infected plant tissue residues, and mechanically through infested tools and equipment during intercultural field operations.

- **Pest and Nematode Vectors:** Plant-parasitic nematodes and soil insects act as physical vectors or predisposing agents by creating feeding wounds on roots and corms that facilitate bacterial entry.

Favorable Environmental Conditions and Aggravating Factors

- **High Soil Temperatures:** During hot summer periods, elevated soil temperatures scorch and injure pseudostem tissues in direct contact with the ground, creating entry wounds for *Erwinia*.
- **Moisture and Rainfall:** Hot, humid, and damp weather accompanied by heavy rainfall accelerates bacterial multiplication. Free surface water or soil saturation is essential for bacterial motility and invasion.
- **Agronomic Triggers:** Poor soil drainage, planting during peak rainy seasons, and using small suckers (weighing under 500 g) significantly increase disease susceptibility.

Damage Erwinia Rhizome Rot causes severe establishment failure in newly planted fields, death of young suckers, complete pseudostem collapse, and substantial population loss in commercial banana orchards.

Integrated Disease Management (IDM)

1. Cultural Practices, Field Preparation, and Sanitation

- **Soil Drainage and Conditioning:** Ensure efficient field drainage and proper soil aeration to prevent waterlogging around plant bases.
- **Planting Window and Material Selection:** Avoid planting during peak monsoon/rainy months. Select large, vigorous suckers weighing more than 500 g. Use rhizomes featuring dead central buds and active lateral buds to minimize disease emergence.
- **Field Hygiene:** Promptly uproot and destroy infected suckers and dead mats. Remove all crop residues and plant debris after harvest to reduce residual soil inoculum.
- **Crop Rotation:** Rotate heavily affected fields with non-host crops such as soybean, forage legumes, or small grains. Avoid intercropping or rotating banana with highly susceptible host crops like onions, cucurbits, or other solanaceous vegetables.
- **Wound Prevention:** Exercise caution during weeding and earthing-up to prevent mechanical injuries to corms and lower pseudostems. Control soil-borne nematodes and insect pests to minimize root wounding.

2. Chemical Drenching and Disinfection Treatments

- **Rhizome / Sucker Drenching:** Drench suckers with 2% Bleaching Powder solution at the time of planting, followed by a second drenching 3 months after planting to eliminate soil-borne bacteria around the root zone.
- **Soil Drenching Options:** Drench the soil surrounding infected or adjacent mats with:
  - 10% Sodium Hypochlorite solution, or
  - Bleaching Powder @ 10 g/L water per plant (or 2% Bleaching Powder solution).

Sources Cited

1. *Banana_disease_1.pdf* (Section: Head Rot, p. 3).
2. *Banana Expert System* (URL; Section: Overcoming the Erwinia rot).
3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.5: Tip over / Rhizome rot, pp. 34–36).

1.5 Banana Bract Mosaic Virus (BBMV / Kokkan Disease)

Causal Agent, Taxonomy, and Susceptible Cultivars Banana Bract Mosaic Disease (commonly known as "Kokkan disease") is caused by the *Banana bract mosaic virus* (BBMV), a member of the *Potyvirus* group12. The virions are flexuous and filamentous2. The disease represents a major economic constraint across southern Indian banana-growing regions2. Highly susceptible cultivars include Nendran, Robusta, Poovan, Ney Poovan, Rasthali, Red Banana, Monthan, and Karpuravalli2.

Symptoms and Identification Symptoms vary depending on plant stage and temperature conditions:

- **Pseudostem and Foliar Streaking:** The diagnostic vegetative symptom is the presence of prominent spindle-shaped pinkish to reddish streaks on the pseudostem, midrib, peduncle, and young suckers23.
- **Foliar Laminar Mottling and Vein Abnormalities:** Young leaves exhibit yellow-green bands or mottling across the lamina, accompanied by abnormal thickening of leaf veins3.
- **Canopy Orientation ("Traveller's Palm"):** In cultivars like Nendran, leaves assume an abnormal, fan-like orientation resembling a "Traveller's palm"2.
- **Diagnostic Floral Bract Mosaic:** Male floral buds turn dark purple and display conspicuous mosaic patches or mottling on the bracts23.
- **Deep-Seated Internal Necrosis:** When the pseudostem is cut horizontally, internal necrotic discoloration is visible deep within the tissue layers2.
- **Bunch Distortion and "Pencil Kai":** Infected plants produce severely reduced bunches with curved, brittle, pencil-sized fingers (locally referred to as *pencil kai*)2. Severely affected plants fail to produce bunches or die prematurely due to stunted growth and extensive pseudostem necrosis23.

Transmission, Vectors, and Epidemiology

- **Primary Transmission:** The virus is primarily spread over distance and between planting cycles through symptomless, infected daughter suckers and tissue-culture plantlets2.
- **Secondary Vector Transmission:** Transmitted locally in a non-persistent manner by aphid vectors, including *Pentalonia nigronervosa* (banana black aphid), *Aphis gossypii* (cotton aphid), *Aphis craccivora* (cowpea aphid), and *Rhopalosiphum maidis* (corn leaf aphid)2.
- **Seasonal Symptom Prominence:** Symptom expression is most distinct during winter months or when ambient temperatures drop below 30°C, making cool periods optimal for visual mother-plant indexing2.

Damage BBMV causes severe stunting, complete bunch malformation, production of unmarketable pencil-sized fruits, and plant mortality, leading to major economic losses in commercial plantations23.

Integrated Disease Management (IDM)

1. Domestic Quarantine, Mother Block Indexing, and Tissue Culture

- Enforce domestic quarantine restrictions to limit the movement of suckers and planting material from endemic southern states to disease-free areas2.
- Establish virus-free mother blocks protected inside vector-proof net houses for sucker extraction2.
- Perform mandatory virus indexing on mother plants and micropropagation explants before mass multiplication using sensitive diagnostic assays such as ELISA or DIBA with reliable antiserum2.
- Select mother plants visually during winter when maximum temperatures are below 30°C to ensure symptom clarity2.

2. Field Sanitation and Eradication

- Inspect plantations regularly and immediately uproot, chop, and burn infected plants along with their underground rhizome systems upon first detection23.
- Use exclusively certified, virus-free tissue-cultured plantlets or indexed suckers for establishing new plantings2.

3. Intercropping Restrictions and Weeding

- Avoid intercropping or growing cucurbitaceous plants (e.g., cucumber, pumpkin, ash gourd, ridge gourd) or solanaceous crops in or around banana fields, as they serve as alternate hosts for viral vectors23.
- Keep fields and surrounding field margins clean and free of weeds that harbor aphids2.

4. Vector Control

- Suppress aphid vector populations by applying foliar sprays of Neemkosol @ 2.5 mL/L or Monocrotophos @ 2 mL/L2.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Banana Bract Mosaic Virus, p. 3).
2. *Banana Expert System* (URL; Section: Do's and Don'ts – Surrounding cultivation).
3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.7: Banana bract mosaic disease, p. 37).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 13: Banana Bract Mosaic, pp. 11–13).

1.6 Infectious Chlorosis / Cucumber Mosaic Virus (CMV)

Causal Agent, Taxonomy, and Vector Host Range Infectious Chlorosis (commonly referred to as Banana Mosaic Disease) is caused by the *Cucumber mosaic virus* (CMV), a member of the *Cucumovirus* group. The virion is an isometric, linear, positive-sense single-stranded RNA (ssRNA) virus. The pathogen possesses an exceptionally wide host range encompassing numerous weed species, vegetable crops, and cultivated ornamentals.

Symptoms and Disease Identification

- **Foliar Mosaic and Chlorotic Streaking:** Primary symptoms appear on young emerging leaves as light green to bright yellow discontinuous linear streaks, patches, or bands extending from the midrib to the leaf margins, creating a distinct mottled or mosaic pattern across the lamina.
- **Foliar Deformities and Stunting:** Infected leaves are narrower and smaller than normal, exhibiting upward margin rolling, leaf twisting, and a rigid, upright stance. Newly formed leaves become crowded at the crown.
- **Dwarfing and Lack of Yield:** Primary infected plants lag severely behind in growth, remain stunted, and fail to produce bunches, remaining in the field solely as virus reservoirs.
- **Heart Rot Phase:** In advanced stage infection, rotting of the heart leaf and central pseudostem tissue occurs, causing young suckers to turn brown, dry out, and die (referred to as the "heart rot" stage).

Transmission, Vectors, and Epidemiology

- **Primary Transmission:** The virus spreads primarily through the movement and planting of infected daughter suckers or rhizome bits taken from diseased mother mats.

- **Secondary Vector Transmission:** Transmitted locally in a non-persistent manner by aphid vectors, specifically the melon aphid (*Aphis gossypii*) and corn leaf aphid (*Rhopalosiphum maidis / Aphids maidis*).
- **Weed and Crop Reservoirs:** Alternate weed hosts (including *Cyperus iria*, *Commelina benghalensis*, *Digitaria* spp., *Amaranthus* spp., *Dactyloctenium aegyptium*, *Chloris barbata*, and *Eragrostis zeylenica*) as well as susceptible intercrops harbor the virus and aphid vectors during off-seasons.

Damage Infectious Chlorosis causes severe plant stunting, total yield loss in early-infected plants, pseudostem heart rot, and persistent orchard degradation.

Integrated Disease Management (IDM)

1. Clean Planting Material and Mother Block Indexing

- Plant exclusively certified, virus-indexed tissue-cultured plantlets or disease-free suckers obtained from verified virus-free mother plants.
- Do not harvest or plant suckers from infected clumps or diseased plantations.

2. Field Hygiene, Weeding, and Eradication

- Inspect fields regularly to detect early mosaic symptoms.
- Uproot, rogue, and burn diseased plants along with their rhizome systems immediately upon detection.
- Maintain plantations and surrounding field borders strictly free of weeds (especially *Commelina benghalensis*, *Cyperus iria*, *Amaranthus* spp., and wild grasses) that harbor the virus during off-seasons.

3. Intercropping Restrictions

- Avoid growing or intercropping susceptible host plants—such as cucurbits (pumpkin, cucumber, ash gourd, ridge gourd, smooth gourd), solanaceous crops (brinjal, tomato, chili), and legumes (beans, moong)—within or adjacent to banana fields.

4. Nutrition and Crop Management Schedule

- Implement a high-input management schedule: apply FYM @ 10 kg/plant at planting and subsequently at 3-month intervals, neem cake @ 1 kg/plant, and balanced NPK fertilizers (200 g N, 40 g P, 200 g K per plant).
- Perform four manual weedings at 2-month intervals up to 8 months after planting.

5. Vector (Aphid) Control

- Apply foliar sprays of systemic insecticides at 3 to 4-week intervals to keep aphid populations suppressed:
  - Dimethoate 30 EC @ 1 mL/L (or 2 mL/L).
  - Methyl demeton / Oxy-demeton methyl 25 EC @ 2 mL/L.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Mosaic Virus, p. 3).
2. *Banana Expert System* (URL; Section: Do's and Don'ts – Surrounding cultivation).
3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.9: Infectious Chlorosis / Cucumber mosaic virus, pp. 38–40).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 13: Preventive strategies for viral diseases, p. 17).

1.7 Banana Streak Virus (BSV) / Banana Streak Disease

Causal Agent, Taxonomy, and Susceptible Cultivars Banana Streak Disease is caused by the *Banana streak virus* (BSV), a badnavirus. It poses a major threat to commercial banana cultivation, particularly affecting cultivars such as Poovan, Grand Nain, Robusta, and cultivars within the Cavendish subgroup (AAA, e.g., 'Poyo').

Symptoms and Disease Identification Symptom expression is erratically distributed across individual leaves and among leaves on the same plant:

- **Foliar Streaking and Progressive Necrosis:** Initial symptoms appear as small, golden-yellow chlorotic dots or flecks that elongate into continuous streaks parallel to leaf veins. Over time, chlorotic streaks turn progressively necrotic, imparting a distinct black-streaked or blackish appearance on mature leaf laminae.
- **Petiole, Midrib, and Stem Necrosis:** Necrotic streaks also develop along the petioles, midribs, and outer pseudostem sheaths.
- **Leaf Stripping:** Prominent leaf stripping symptoms are frequently observed in cultivars Poovan, Grand Nain, and Robusta.
- **Diagnostic Distinction from CMV:** Unlike Banana Mosaic Disease (CMV), Banana Streak Disease does **not** produce the characteristic midrib mosaic symptom.
- **Bunch Malformation and Fruit Abnormalities:** Infected plants exhibit severe bunch choking, bunch abortion, seediness in fruits, smaller bunches, and distorted fingers with abnormally thin peels.

Transmission, Vectors, and Genomic Integration

- **Primary Spread:** The primary mode of transmission across regions is through infected, vegetatively propagated planting material (suckers and rhizome bits).
- **Secondary Vector Transmission:** Transmitted locally in a semi-persistent manner by mealybug vectors, including *Planococcus citri*, *Planococcus ficus*, *Dysmicoccus brevipes*, *Pseudococcus longispinus*, *Saccharicoccus sacchari*, and *Ferrisia virgata*.

- **Non-Transmission by Sap/Tools:** The virus is **not** transmitted mechanically through sap or by tools used during field cultivation.
- **Endogenous Integration and Tissue Culture Risks:** Standard shoot tip culture does not eliminate BSV from infected vegetative tissue. Furthermore, tissue culture stress can trigger the activation and episomal expresson of integrated endogenous viral sequences present within the host plant genome, giving rise to de novo viral infections.

Favorable Environmental Conditions and Symptom Expression Temperature plays a critical role in controlling symptom expression. Symptom development fluctuates seasonally; maintaining plants at 22°C in a temperature-controlled glasshouse triggers reliable symptom expression, making cool conditions ideal for diagnostic screening.

Damage and Commercial Impact BSV causes severe growth stunting, bunch choking, total bunch abortion, fruit distortion, thin fruit skins, and seediness, rendering affected produce completely unmarketable and causing heavy economic losses in Poovan and Cavendish plantations.

Integrated Disease Management (IDM)

1. Clean Planting Material, Indexing, and Glasshouse Screening

- Establish new plantations using exclusively certified virus-free planting material sourced from disease-free mother blocks.
- Perform mandatory molecular and visual virus indexing before mass propagation.
- Maintain tissue culture mother plants in temperature-controlled glasshouses at 22°C for 2 to 3 months prior to explant extraction to induce latent symptom expression and allow reliable indexing.

2. Field Monitoring, Sanitation, and Eradication

- Inspect plantations regularly for early chlorotic or necrotic leaf streaking.
- Promptly uproot, remove, and bury or burn infected plants to eliminate field inoculum reservoirs. Replace destroyed mats with certified disease-free plants.

3. Vector Management and Quarantine

- Monitor and suppress mealybug populations (*Planococcus* spp., *Dysmicoccus* spp.) around the pseudostem bases and leaf axils using recommended systemic chemical interventions or bio-agents.
- Enforce domestic quarantine measures to restrict the movement of unindexed suckers from infected zones.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Banana Streak Virus, p. 3).
2. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.8: Banana streak disease, p. 38).
3. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 14: Banana streak disease, pp. 13–15).

1.8 Banana Anthracnose (*Colletotrichum musae / Gloeosporium musae*)

Causal Agent and Taxonomy Banana Anthracnose is a major pre- and post-harvest fungal disease caused by *Colletotrichum musae* (Berk. & Curt.) v. Arx (syn. *Gloeosporium musae / Gloeosporium musarum*). The disease attacks banana plants at all stages of development, affecting flowers, green fruit in the field, and mature fruits during storage and ripening.

Symptoms and Disease Identification Infection manifests through two distinct expression phases:

- **Latent Field Infection on Green Fruit:** Infection typically initiates during the green fruit stage in the field via flower remnants or peel injuries, remaining latent until the fruit undergoes post-harvest ripening. Early field symptoms appear as small, black, circular, or blister-like spots and streaks on flowers, fruit skin, and distal ends of banana heads (hands).
- **Mature Lesions on Green Skin:** Lesions expand into dark brown to black, diamond-shaped spots (measuring up to 8 cm×3 cm) or circular lesions that become sunken and are delimited from healthy green tissue by a distinct pale margin. The internal pulp directly beneath the lesion becomes degraded.
- **Post-Harvest Ripening Symptoms:** As fruits ripen, numerous small, dark, circular spots develop rapidly, enlarge, coalesce, and become deeply sunken.
- **Spore Masses and Fruit Decay:** Upon maturity of lesions, mass production of conidia forms a characteristic pinkish to salmon-pink slimy spore coat over the affected patches. In severe cases, the entire fruit and bunch turn black, shrivel, and suffer premature fruit drop or complete rot.

Disease Cycle, Primary Inoculum, and Spread

- **Primary Inoculum:** The fungus survives and sporulates primarily in infected banana trash, dried hanging leaf residues, and decaying floral parts around the plantation.
- **Dissemination and Vector Spread:** Conidia are dispersed through water droplets, rain splashes, air currents, and insects (such as bees and thrips) that frequently visit banana flowers.

- **Entry via Physical Wounds:** Mechanical injuries, skin abrasions, and bruising during harvesting, field handling, and transport significantly increase fruit susceptibility and trigger non-latent post-harvest infection.

Favorable Environmental Conditions

- **Temperature:** Disease incidence and severity peak at high ambient temperatures between 30°C and 35°C.
- **Relative Humidity and Rainfall:** Relative humidity above 85% accompanied by persistent rainfall or high atmospheric moisture favors rapid spore germination and infection.
- **Seasonal Peak:** Outbreaks are most severe during summer and monsoon months (June through September). Lack of a proper prophylactic spray schedule and accumulation of banana trash aggravate disease intensity.

Damage and Post-Harvest Losses Anthracnose causes extensive commercial degradation of harvested bunches. Latent field infections remain invisible during harvest but cause rapid fruit blackening, shriveling, pulp breakdown, and unmarketability during transit, cold storage, and artificial ripening.

Integrated Disease Management (IDM)

1. Field Sanitation and Cultural Management

- Maintain strict plantation hygiene by regularly removing, burning, or burying dead hanging leaves, infected plant residues, and leaf trash to minimize inoculum load.
- Keep fields free of weeds and ensure efficient soil drainage.
- Apply balanced fertilizer applications based on soil testing to improve fruit peel resistance against fungal penetration.
- Harvest banana bunches at the correct stage of maturity, ensuring fruit is free from field infections prior to transport, storage, and ripening.

2. Bunch Sleeving and Physical Protection

- Sleeve developing bunches with non-perforated or ventilated polyethylene bags as soon as possible after the removal of the male flower bud to shield fruits from airborne spores and insect vectors.

3. Careful Harvesting, Handling, and Cold Chain Sanitation

- Minimize fruit bruising, skin cuts, and mechanical impact injuries during harvesting, field transport, and packing house operations.
- Disinfect handling equipment, wash tanks, and packing house facilities thoroughly.
- Maintain prompt refrigeration and cool fruit down to 14°C in cold storage to arrest fungal development during transit.

4. Botanical Post-Harvest Treatment

- Dip harvested fruits in a water extract of *Solanum torvum* @ 50% concentration, which effectively reduces Anthracnose incidence by over 90%.

5. Prophylactic Chemical Fungicides

- **Pre-Harvest Spray Schedule:** Apply pre-harvest sprays during humid or rainy weather (starting 15 days prior to harvest) using one of the following fungicide formulations:
  - Copper Oxychloride 50% WP @ 2.5 g/L (or 2.5 g/mL).
  - Chlorothalonil @ 0.2% (1.5 mL/L).
  - Carbendazim / Bavistin @ 1% or 2 g/L.
  - Prochloraz @ 1 mL/L.
  - Thiophanate Methyl or Bitertanol @ 2 g/L.
- **Spray Frequency:** Apply up to 4 sprays at 15-day intervals as needed during high-risk weather.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Anthracnose, pp. 2–3).
2. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.3: Anthracnose, pp. 32–33).
3. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 10: Leaf Spot / Anthracnose, pp. 14–15; Section 15: IPM Schedule, p. 18).

1.9 Cigar End Tip Rot (*Verticillium theobromae*)

Causal Agents and Taxonomy Cigar End Tip Rot (also termed Cigar End Rot) is a fungal fruit disease affecting developing banana bunches.

- **Primary Pathogen:** *Verticillium theobromae* (Turc.) Mason & Hughes.
- **Associated Fungal Complex:** *Banana_disease_1.pdf* (p. 3) also identifies *Trachysphaera fructigena* and *Gloeosporium musarum* as co-involved pathogens in cigar end necrosis.

Symptoms and Disease Identification Infection manifests during early finger development and progresses along the fruit:

- **Distal Necrosis:** A dark, blackish necrosis initiates at the persistent perianth or flower end and creeps slowly along the distal tips of immature fingers.
- **Diagnostic "Cigar Ash" Mycelia:** As the infected tissue shrivels and dries, it becomes covered with a powdery coat of greyish fungal conidia and mycelia, closely resembling the grey ash of a burnt cigar.

- **Fruit Skin and Pulp Decay:** The skin of the affected fruit becomes shrunken, folded, and black-pitted. The internal pulp undergoes a characteristic dry rot that spreads up to 2c̆m from the tip, leaving the pulp mummified.
- **Early Bunch Attack:** If infection occurs on newly emerging fingers, the rot can consume the entire finger, causing premature fruit drying and drop.

Epidemiology, Inoculum Sources, and Spread

- **Inoculum Reservoirs:** The pathogen survives saprophytically in leaf trash, decaying banana foliage, and persistent, drying floral parts (petals, styles, and perianth).
- **Dissemination:** Conidia are air-borne and disseminated by wind currents and rain splashes onto open flowers and young fingers.

Favorable Conditions and Commercial Damage Disease incidence peaks during warm summer months. High relative humidity and the accumulation of senescent floral structures around developing hands accelerate spore germination. In severe outbreaks, affected hands become unmarketable due to unsightly black distal necrosis and dry pulp decay.

Integrated Disease Management (IDM)

1. Cultural Practices and Sanitation

- **Deflowering / Perianth Removal:** Manually remove the persistent style, pistil, and perianth from fully emerged fingers by hand 8 to 10 days after bunch formation (just after flowering).
- **Field Hygiene:** Maintain orchard sanitation by removing dry, hanging leaves and trash to improve canopy aeration. Keep packing and ripening sheds clean.

2. Bunch Sleeving / Bagging

- Bag or sleeve developing bunches with polyethylene covers (e.g., 100-gauge vented bags) immediately after the emergence of hands and removal of the male flower bud to shield fingers from airborne conidia.

3. Fungicidal Spray Interventions Apply prophylactic sprays to developing bunches as soon as fingers form using one of the following treatments:

- Mancozeb 75% WP @ 1.5–2 kg/ha (or Dithane M-45 @ 0.1% / 2.5 mL/L water).
- Topsin M @ 0.1% (1 g/L).
- Benlate @ 0.5%.
- Thiophanate Methyl or Bitertanol @ 1 g/L, or Chlorothalonil.

4. Post-Harvest Handling and Cold Chain

- Avoid skin abrasions and bruising during harvesting and transit.
- Disinfect handling facilities and promptly cool harvested fruit down to 14°C in cold storage to inhibit post-harvest fungal expansion.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Cigar End Tip Rot, p. 3).
2. *Banana Expert System* (URL; Sections: Eighth Month, Do's and Don'ts).
3. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.4: Cigar end rot, pp. 33–34).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 11: Cigar-end rot, pp. 10–11).

1.10 Crown Rot, Stem-End Rot, and Black Tip Decay Complex

Causal Agents and Species Complex Post-harvest crown and stem-end rots are caused by a complex of wound-invading opportunistic fungi acting individually or synergistically:

- **Crown Rot Complex:** Caused by *Colletotrichum musae*, *Fusarium* spp., *Verticillium theobromae*, *Botryodiplodia theobromae*, and *Nigrospora sphaerica*.
- **Stem-End Rot:** Caused primarily by *Thielaviopsis paradoxa*.
- **Finger Tip / Black Tip Rot:** Caused by *Botryodiplodia theobromae* Pat.

Symptoms and Disease Identification

- **Crown Rot:** Initial symptom is a dark blackening of the cut crown surface after hand dehanding. The rot advances through pedicels into the fruit pulp, causing soft decay of the crown-finger connection and leading to fingers detaching ("dropping off") from the hand during handling.
- **Stem-End Rot:** Fungus invades through cut pedicels or bunch stems. Invaded fruit flesh becomes soft, dark, and water-soaked, quickly spreading through the central pulp.
- **Finger Tip / Black Tip Rot:** Characterized by a surface mass of greyish-black mycelium starting from persistent perianths or stem ends. Infection progresses to turn fruit skin black, soft, and encrusted with dark pycnidial fruiting bodies, while internal pulp degrades into a black watery mass.

Entry Points and Post-Harvest Spread The fungi infect harvested bunches through mechanical cuts, bruised tissue, and fresh crown cut surfaces created during dehanding and field trimming. High packing house humidity and elevated storage temperatures accelerate hyphal invasion.

Post-Harvest Management and Chemical Controls

- **Sanitation and Careful Handling:** Disinfect wash tanks and handling facilities. Handle bunches with care to minimize cuts, scratches, and impact bruises. Trim affected portions of hands carefully.
- **Fungicidal Crown Treatments:** Dip harvested bunches or hands in Thiabendazole or Benomyl solutions, or pack hands using cellulose pads impregnated with fungicides.
- **Pre- and Post-Harvest Dips / Sprays:** Apply pre- or post-harvest sprays/dips of Carbendazim, Bitertanol, Thiophanate Methyl, Prochloraz @ 1 g/L, or Rovral @ 2 g/L. Alternatively, apply 1% Bordeaux mixture or Dithane M-45 @ 0.3%.
- **Hot Water Treatment:** Dip hands in hot water @ 50°C for 5 minutes to control stem-end rot fungi.
- **Fruit Waxing and Cold Storage:** Coat fruit with a wax emulsion containing benomyl to slow fruit ripening, limit moisture loss, and prevent rot development. Maintain refrigeration at 10°C–14°C during cold transit.

Sources Cited

1. *Banana_disease_1.pdf* (Sections: Crown Rot, Stem-end Rot, p. 3).
2. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 12: Finger tip or black tip rots, p. 11).

1.11 Pseudostem Heart Rot (*Botryodiplodia* sp., *Gloeosporium* sp., and *Fusarium* sp.)

Causal Agents and Associated Fungal Complex Pseudostem Heart Rot is a fungal disease complex affecting the growing crown and heart leaves of banana. The disease is attributed to infection by one or a combination of fungal pathogens, primarily *Botryodiplodia* species, *Gloeosporium* species, and *Fusarium* species.

Symptoms and Disease Identification

- **Foliar Lamina Decay:** The earliest visual indication of heart rot is the appearance of emerging central heart leaves with portions of the leaf lamina missing, distorted, or actively decaying.
- **Crown Yellowing and Mortality:** In severe cases, the innermost young leaves at the crown turn yellow, progress to dark brown, and ultimately die.
- **Total Canopy Death:** In advanced or unmanaged infections, necrosis spreads through the growing point, causing all leaves and the entire plant to collapse and die.

Favorable Conditions and Aggravating Factors Heart rot incidence is aggravated by poor soil drainage, waterlogging around the crown base, inadequate field aeration resulting from high-density or improper plant spacing, and high atmospheric humidity.

Prevention and Management

- **Cultural Practices and Field Sanitation:** Adopt strict field hygiene by clearing decayed crop residues. Provide adequate field drainage to prevent moisture accumulation around the crown, and ensure proper plant spacing during field layout.
- **Fungicidal Sprays:** Prevent disease expansion and secondary spread by applying foliar sprays of protective fungicides:
  - Captan, or
  - Dithane M-45 (Mancozeb), or
  - Dithane Z-78 (Zineb).

Sources Cited

1. *Banana_disease_1.pdf* (Section: Pseudostem Heart Rot, p. 3).

1.12 Bacterial Wilt / Moko Disease (*Pseudomonas solanacearum* / *Ralstonia solanacearum*)

Causal Agent and Host Susceptibility Bacterial Wilt (commonly known as Moko Disease) is caused by the bacterium *Pseudomonas solanacearum* (syn. *Ralstonia solanacearum*). Young banana plants are particularly vulnerable to severe infection.

Symptoms and Identification

- **Foliar Chlorosis and Leaf Petiole Collapse:** Early infection is marked by yellowish discoloration of the inner leaf lamina adjacent to the petiole. The leaf collapses at the junction between the lamina and petiole.
- **Rapid Foliar Wilting:** Within one week of initial chlorosis, the majority of the canopy exhibits severe wilting symptoms.
- **Diagnostic Finger Yellowing:** The presence of premature yellow fingers within an otherwise green, immature bunch serves as a key diagnostic field indicator of Moko disease.
- **Diagnostic Sucker Stunting and Blackening:** The most characteristic symptoms occur on young ratoon suckers that have been cut back once and started regrowth; emerging suckers become severely stunted, blackened, and produce tender leaves that rapidly turn yellow and necrotic.

Infection Spread and Insect Transmission

- **Insect Vector Transmission:** Foraging insects act as mechanical vectors by carrying bacterial inoculum to open male flowers (navels).
- **Tool Transmission:** The bacterium is readily spread mechanically between mats on contaminated pruning knives, machetes, and dehanding tools.

Integrated Disease Management (IDM)

- **Early Detection and Destruction:** Perform regular orchard monitoring. Promptly uproot and destroy all suspected plants and their entire root mats to halt field spread.
- **Tool Disinfection:** Disinfect all cutting tools, knives, and agricultural implements with formaldehyde or spirit before and after use in each plot.
- **Denavelling (Male Flower Removal):** Remove the male flower bud (denavelling) immediately after the last female hand has emerged to eliminate the primary flower entry point for insect-borne bacterial inoculum.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Bacterial Wilt or Moko Disease, p. 3).

1.13 Cordana Leaf Spot / Diamond Leaf Spot (*Cordana musae* / *Neocordana musae*)

Causal Agents and Taxonomy Cordana Leaf Spot (also known as Banana Diamond Leaf Spot or Leaf Blotch) is caused primarily by the fungus *Cordana musae* (Zimm.) Höhn. (syn. *Neocordana musae*). In cooler banana-growing climates, a related species, *Cordana johnstonii* (*Neocordana johnstonii*), is also reported. The pathogen widely acts as an opportunistic or secondary invader, rapidly colonizing leaf tissues damaged by physical injury, environmental stress, or primary lesions caused by other fungi (such as *Mycosphaerella musicola*).

Symptoms and Identification Symptoms develop predominantly on mature, lower foliage:

- **Foliar Lesion Morphology:** Oval, eye-shaped, or diamond-shaped (fusiform) pale brown to greyish-brown necrotic spots (measuring up to 10c̆m in length) appear on the upper leaf surface.
- **Concentric Zonation and Yellow Halo:** Lesions feature distinct concentric rings or target-like zonation within the necrotic center, bordered by a dark brown band and surrounded by a conspicuous, bright yellow to orange halo.
- **Marginal Bands and Stripe Formation:** Infections frequently initiate near the leaf margins, enlarging into long, pale brown necrotic bands or stripes that extend along leaf veins toward the midrib.
- **Coalescence and Leaf Blight:** Multiple spots coalesce into extensive necrotic patches, causing the entire leaf lamina to turn brown, desiccate, and die prematurely.
- **Diagnostic Fungal Sporulation:** Under humid conditions, abundant greyish-brown, velvety/hairy fungal conidia and conidiophores are produced on the underside of infected leaves.

- **Interspecific Species Differences:** Lesions produced by *C. musae* are significantly larger and more elongated than those caused by *C. johnstonii* (which produces smaller spots, approximately 10 mm × 15 mm).

Infection Cycle, Spread, and Favorable Environmental Conditions

- **Dissemination and Entry:** Fungal conidia are dispersed locally by wind currents and water splash (rain or sprinkler irrigation). Pathogen entry occurs through natural leaf openings, physical/mechanical leaf damage, or pre-existing lesions caused by Sigatoka or insect pests.
- **Environmental Triggers:** Disease development accelerates under hot, humid weather with frequent rains, heavy dew, and high canopy humidity.
- **Aggravating Stress Factors:** Plants are heightened to infection when suffering from environmental stress (such as waterlogging or drought), nutrient imbalances (e.g., excessive nitrogen fertilization), mutual shading in high-density plantings, or lack of proper weed control.

Damage and Commercial Severity While widely distributed across tropical banana production zones, Cordana leaf spot is generally classified as a secondary leaf disease. However, in unsprayed plantations, high-density plots, or on susceptible local cultivars under humid conditions, severe outbreaks destroy significant functional green leaf area, impairing plant photosynthesis and yield. Cavendish cultivars generally exhibit higher field tolerance compared to plantains and local cultivars.

Integrated Disease Management (IDM)

1. Cultural Practices and Field Sanitation

- **Planting Density and Canopy Ventilation:** Maintain proper plant spacing (e.g., 2.5 m × 2.5 m to 3.5 m × 3.5 m / 8 ft × 8 ft to 12 ft × 12 ft) to reduce canopy crowding, prevent mutual leaf contact, and enhance air circulation.
- **Deleafing and Sanitation:** Systematically remove, burn, or destroy infected, old, or dried hanging leaves to eliminate field spore reservoirs. Maintain clean, weed-free plantations.
- **Irrigation Management:** Prefer drip/trickle irrigation over sprinkler or overhead irrigation to minimize relative humidity in the canopy and reduce leaf wetness duration.
- **Balanced Nutrition:** Apply balanced NPK fertilizers based on soil tests, avoiding excessive nitrogen applications that induce succulent, disease-prone leaf growth.

2. Chemical Fungicide Spray Recommendations

- **Incidental Sigatoka Management:** Standard Sigatoka fungicide spray programs generally provide effective concurrent suppression of Cordana leaf spot.

- **Threshold and Protective Sprays:** When disease extends above the 4th leaf from the bottom, apply targeted protective or systemic fungicidal sprays:
  - **Contact Fungicides:** Mancozeb 75% WG / WP @ 2 g/L (0.2–0.4%), Chlorothalonil 500 g/L SC @ 3 mL/L, or organic copper formulations (e.g., 1% Bordeaux mixture or Copper Oxychloride @ 2–4 g/L).
  - **Systemic Fungicides:** Tebuconazole 250 g/L EW @ 0.35 mL/L or Propiconazole 250 g/L EC @ 0.35 mL/L (or registered premixes such as Luna® Experience or Scala®).
- **Spray Coverage:** Ensure sprays thoroughly cover upper and lower leaf surfaces, particularly in the upper canopy. Alternate contact and systemic chemistry to prevent fungicide resistance.